

Jongsun Park

professor

基本信息

姓名: Jongsun Park

所属院校: 高丽大学

地区: South Korea

学科领域

集成电路设计与集成系统

电子与计算机工程

人工智能

研究领域

低功耗电路与能量管理

存储器设计与架构

低功耗设计与能耗优化

深度学习与神经网络

计算机体系结构

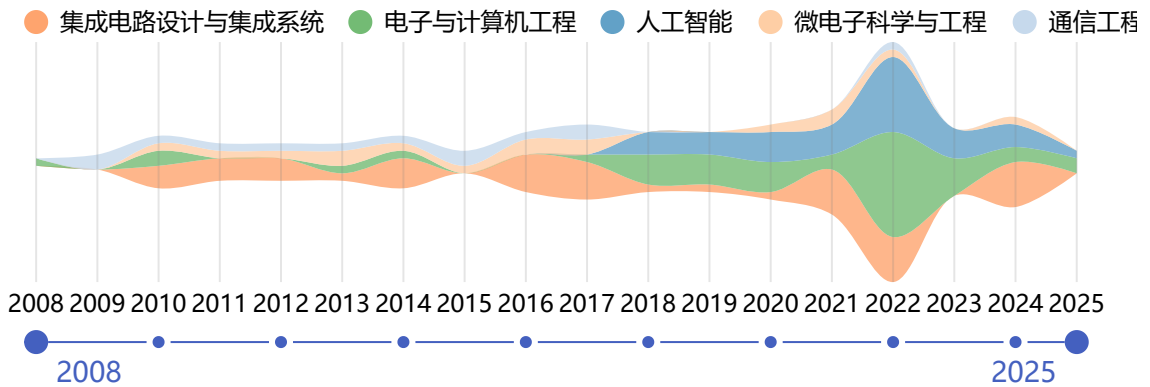
数字集成电路设计

嵌入式系统与硬件设计

个人简介

Jongsun Park is a professor at the School of Electrical Engineering. His research activity spans from 2000 - 2024, with an h - index of 2861 and 24 citations. He is actively involved in topics like Energy Engineering, Low Power, and Energy Efficient. His research output includes 97 articles, 81 conference contributions, 4 conference articles, 3 papers, 3 editorials, and 1 review article. In 2024, he has multiple publications in top journals such as IEEE Journal of Solid - State Circuits and IEEE Transactions on Computers, covering areas like SRAM design, STT - MRAM architecture, and spiking neural networks.

近年发表的论文



Fault Bounding On-Die BCH Codes for Improving Reliability of System ECC

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kang, Seongyoon; Shin, Chaehyeon; Park, Jongsun

被引用次数: 0

2025

A Design Framework of Heterogeneous Approximate DCIM-Based Accelerator for Energy-Efficient NN Processing

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Lee, Kyeongho; Lee, Hyeyeong; Park, Jongsun

被引用次数: 0

2025

Highly Reliable Magnetic Memory-Based Physical Unclonable Functions

ACS NANO

Kang, Jaimin; Han, Donghyeon; Lee, Kyungchul; Ko, San; Koh, Daekyu; Park, Chando; Ahn, Jaesoo; Yu, Minrui; Pakala, Mahendra; Noh, Sujung; Lee, Hansaem; Kwon, Joonhyun; Kim, Kab-Jin; Park, Jongsun; Lee, Soogil; Lee, Jisung; Park, Byong-Guk

被引用次数: 2

2024

Large scale integrated IGZO crossbar memristor array based artificial neural architecture for scalable in-memory computing

MATERIALS TODAY NANO

Naqi, Muhammad; Kim, Taehwan; Cho, Yongin; Pujar, Pavan; Park, Jongsun; Kim, Sunkook

被引用次数: 6

2024

Spintronic Artificial Synapses Using Voltage-Controlled Multilevel Magnetic States

ADVANCED ELECTRONIC MATERIALS

Jeong, Jimin; Jang, Yunho; Kang, Min-Gu; Hwang, Seungeon; Park, Jongsun; Park, Byong-Guk

被引用次数: 0

2024

A 28-nm 50.1-TOPS/W P-8T SRAM Compute-In-Memory Macro Design With BL Charge-Sharing-Based In-SRAM DAC/ADC Operations

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Kyeongho; Kim, Joonhyung; Park, Jongsun

被引用次数: 3

2024

SOT-MRAM-Based LUT Cell Design for Area- and Energy-Efficient FPGA

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Kim, Dongsu; Kim, Taehwan; Jang, Yunho; Park, Jongsun

被引用次数: 0

2024

STT-MRAM-based Near-Memory Computing Architecture with Read Scheme and Dataflow Co-Design for High-Throughput and Energy-Efficiency

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Jang, Yunho; Kim, Yeseul; Park, Jongsun

被引用次数: 0

2024

iSPADE: End-to-end Sparse Architecture for Dense DNN Acceleration via Inverted-bit Representation

PROCEEDINGS OF THE 29TH ACM/IEEE INTERNATIONAL SYMPOSIUM ON LOW POWER ELECTRONICS AND DESIGN, ISLPED 2024

Kim, Dongjun; Cho, Han; Park, Jongsun

被引用次数: 0

2024

HeNCoG: A Heterogeneous Near-memory Computing Architecture for Energy Efficient GCN Acceleration

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Hwang, Seung-Eon; Song, Duyeong; Park, Jongsun

被引用次数: 0

2024

Optical Verification of Physically Unclonable Function Devices Based on Spin-Orbit Torque Switching

ADVANCED ELECTRONIC MATERIALS

Lee, Jeong Kyu; Lee, Jiyoung; Yoon, Seok In; Lee, Min Hyeok; Lee, Jin Seo; Jang, Yunho; Kim, Dae-Yun; Choe, Sug-Bong; Park, Jongsun; Kim, Young Keun

被引用次数: 5

2023

Improved Robustness against Magnetic Field in Spin-Orbit-Torque-Based Physical Unclonable Functions through Write-Back Operation

ADVANCED ELECTRONIC MATERIALS

Koh, Daekyu; Kang, Jaimin; Kim, Taehwan; Lee, Jisung; Noh, Sujung; Lee, Hansaem; Kwon, JoonHyun; Lee, Soogil; Park, Jongsun; Park, Byong-Guk

被引用次数: 2

2023

Low Complexity Gradient Computation Techniques to Accelerate Deep Neural Network Training

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Shin, Dongyeob; Kim, Geonho; Jo, Joongho; Park, Jongsun

被引用次数: 2

2023

LoCoExNet: Low-Cost Early Exit Network for Energy Efficient CNN Accelerator Design

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Jo, Joongho; Kim, Geonho; Kim, Seungtae; Park, Jongsun

被引用次数: 1

2023

A 2941-TOPS/W Charge-Domain 10T SRAM Compute-in-Memory for Ternary Neural Network

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Cheon, Sungsoo; Lee, Kyeongho; Park, Jongsun

被引用次数: 9

2023

Low Area and Low Power Threshold Implementation Design Technique for AES S-Box

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Song, Junhyun; Lee, Kyeongho; Park, Jongsun

被引用次数: 0

2023

Neuron pruning in temporal domain for energy efficient SNN processor design

FRONTIERS IN NEUROSCIENCE

Lew, Dongwoo; Tang, Hoyoung; Park, Jongsun

被引用次数: 0

2023

Multilevel artificial electronic synaptic device of direct grown robust MoS2 based memristor array for in-memory deep neural network

NPJ 2D MATERIALS AND APPLICATIONS

Naqi, Muhammad; Kang, Min Seok; Liu, Na; Kim, Taehwan; Baek, Seungho; Bala, Arindam; Moon, Changgyun; Park, Jongsun; Kim, Sunkook

被引用次数: 47

2022

SOT-MRAM Digital PIM Architecture With Extended Parallelism in Matrix Multiplication

IEEE TRANSACTIONS ON COMPUTERS

Kim, Taehwan; Jang, Yunho; Kang, Min-Gu; Park, Byong-Guk; Lee, Kyung-Jin; Park, Jongsun

被引用次数: 10

2022

Stochastic SOT Device Based SNN Architecture for On-Chip Unsupervised STDP Learning

IEEE TRANSACTIONS ON COMPUTERS

Jang, Yunho; Kang, Gyuseong; Kim, Taehwan; Seo, Yeongkyo; Lee, Kyung-Jin; Park, Byong-Guk; Park, Jongsun

被引用次数: 12

2022

A Dual-Domain Dynamic Reference Sensing for Reliable Read Operation in SOT-MRAM

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Kim, Jooyoon; Jang, Yunho; Kim, Taehwan; Park, Jongsun

被引用次数: 9

2022

Area and Energy Efficient Joint 2T SOT-MRAM-Based on Diffusion Region Sharing With Adjacent Cells

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Jang, Yunho; Park, Jongsun

被引用次数: 11

2022

Early Termination Based Training Acceleration for an Energy-Efficient SNN Processor Design

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Choi, Sunghyun; Lew, Dongwoo; Park, Jongsun

被引用次数: 5

2022

A Time-to-first-spike Coding and Conversion Aware Training for Energy-Efficient Deep Spiking Neural Network Processor Design

PROCEEDINGS OF THE 59TH ACM/IEEE DESIGN AUTOMATION CONFERENCE, DAC 2022

Lew, Dongwoo; Lee, Kyungchul; Park, Jongsun

被引用次数: 11

2022

Low-Cost 7T-SRAM Compute-In-Memory Design based on Bit-Line Charge-Sharing based Analog-To-Digital Conversion

2022 IEEE/ACM INTERNATIONAL CONFERENCE ON COMPUTER AIDED DESIGN, ICCAD

Lee, Kyeongho; Kim, Joonhyung; Park, Jongsun

被引用次数: 8

2022

Source-Line Shared SOT-MRAM Cell for Energy Efficient Read Operation

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOC)

Kim, Taehwan; Park, Jongsun

被引用次数: 4

2022

SRAM Bit-line Boosting Circuit for Low Latency and Timing Aware Read Operation

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOC)

Lee, Hyeyeong; Kim, Joonhyung; Park, Jongsun

被引用次数: 1

2022

Bit-Line Decoupled SRAM for Reducing Read Delays in Near Threshold Voltage Operations

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOC)

Park, Hyunchul; Park, Jongsun

被引用次数: 1

2022

Energy-Efficient STT-MRAM based Digital PIM supporting Vertical Computations Using Sense Amplifier

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOC)

Kim, Yeseul; Park, Jongsun

被引用次数: 0

2022

Distributed Accumulation based Energy Efficient STT-MRAM based Digital PIM Architecture

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOC)

Kim, Dongsu; Park, Jongsun

被引用次数: 1

2022

High Detection Rate BCH Code with CRC Code for Memory Application

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Kim, Minseo; Park, Jongsun

被引用次数: 0

2022

Channel-Wise Activation Map Pruning using MaxPool for Reducing Memory Accesses

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Cho, Han; Park, Jongsun

被引用次数: 2

2022

Percentile Clipping based Low Bit-Precision Quantization for Depth Estimation Network

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Hwang, Seungeon; Park, Jongsun

被引用次数: 0

2022

A Layer-wise Training and Pruning Method for Memory Efficient On-chip Learning Hardware

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Lew, Dongwoo; Park, Jongsun

被引用次数: 0

2022

Data Bus Inversion Encoding for Improving the Power Efficiency of SERDES-Containing Data Bus

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Kang, Seongyoon; Park, Jongsun

被引用次数: 0

2022

The Quantitative Comparisons of Analog and Digital SRAM Compute-In-Memories for Deep Neural Network Applications

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Kim, Joonhyung; Park, Jongsun

被引用次数: 2

2022

Clipped Quantization Aware Training for Hardware Friendly Implementation of Image Classification Networks

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Lee, Kyungchul; Park, Jongsun

被引用次数: 1

2022

Class Difficulty based Mixed Precision Quantization for Low Complexity CNN Training

2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)

Jo, Joongho; Park, Jongsun

被引用次数: 3

2022

A time-to-first-spike coding and conversion aware training for energy-efficient deep spiking neural network processor design

Design Automation Conference

Dongwoo Lew; Kyunchul Lee; Jongsun Park

被引用次数: 0

2022

Shared-Write-Channel-Based Device for High-Density Spin-Orbit-Torque Magnetic Random-Access Memory

PHYSICAL REVIEW APPLIED

Mishra, Rahul; Kim, Taehwan; Park, Jongsun; Yang, Hyunsoo

被引用次数: 10

2021

Exploiting Retraining-Based Mixed-Precision Quantization for Low-Cost DNN Accelerator Design

IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS

Kim, Nahsung; Shin, Dongyeob; Choi, Wonseok; Kim, Geonho; Park, Jongsun

被引用次数: 18

2021

A 65-nm 0.6-fJ/Bit/Search Ternary Content Addressable Memory Using an Adaptive Match-Line Discharge

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lee, Kyeongho; Choi, Woong; Park, Jongsun

被引用次数: 4

2021

An Error Compensation Technique for Low-Voltage DNN Accelerators

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ji, Daehan; Shin, Dongyeob; Park, Jongsun

被引用次数: 7

2021

Electric-field control of field-free spin-orbit torque switching via laterally modulated Rashba effect in Pt/Co/AlOx structures

NATURE COMMUNICATIONS

Kang, Min-Gu; Choi, Jong-Guk; Jeong, Jimin; Park, Jae Yeol; Park, Hyeon-Jong; Kim, Taehwan; Lee, Taekhyeon; Kim, Kab-Jin; Kim, Kyoung-Whan; Oh, Jung Hyun; Viet, Duc Duong; Jeong, Jong-Ryul; Yuk, Jong Min; Park, Jongsun; Lee, Kyung-Jin; Park, Byong-Guk

被引用次数: 49

2021

A Charge-Sharing based 8T SRAM In-Memory Computing for Edge DNN Acceleration

Design Automation Conference

Kyeongho Lee; Sungsoo Cheon; Joongho Jo; Woong Choi; Jongsun Park

被引用次数: 0

2021

Low Energy and Error Resilient SOT- MRAM based FPGA LUT Cell

18TH INTERNATIONAL SOC DESIGN CONFERENCE 2021 (ISOC 2021)

Kim, Dongsu; Park, Jongsun

被引用次数: 0

2021

A Charge-domain 10T SRAM based In-Memory-Computing Macro for Low Energy and Highly Accurate DNN inference

18TH INTERNATIONAL SOC DESIGN CONFERENCE 2021 (ISOC 2021)

Kim, Joonhyung; Park, Jongsun

被引用次数: 6

2021

Short Word-Line Pulse with Fast Bit-Line Boosting For High Throughput 6T SRAM-based Compute Inmemory Design

18TH INTERNATIONAL SOC DESIGN CONFERENCE 2021 (ISODC 2021)

Kim, Minseo; Park, Jongsun

被引用次数: 0

2021

Local Bit-line Charge-sharing based Pre-charging SRAM for Near Threshold Voltage Operation

18TH INTERNATIONAL SOC DESIGN CONFERENCE 2021 (ISODC 2021)

Park, Hyunchul; Park, Jongsun

被引用次数: 0

2021

Offset-canceling Sensing Scheme using Feedback for Read Margin Improvement in STT-MRAMs

2021 36TH INTERNATIONAL TECHNICAL CONFERENCE ON CIRCUITS/SYSTEMS, COMPUTERS AND COMMUNICATIONS (ITC-CSCC)

Kim, Dongsu; Kim, Jooyoon; Park, Jongsun

被引用次数: 0

2021

Optimization Techniques for Conversion of Quantization Aware Trained Deep Neural Networks to Lightweight Spiking Neural Networks

2021 36TH INTERNATIONAL TECHNICAL CONFERENCE ON CIRCUITS/SYSTEMS, COMPUTERS AND COMMUNICATIONS (ITC-CSCC)

Lee, Kyungchul; Choi, Sunghyun; Lew, Dongwoo; Park, Jongsun

被引用次数: 0

2021

A Charge-Sharing based 8T SRAM In-Memory Computing for Edge DNN Acceleration

2021 58TH ACM/IEEE DESIGN AUTOMATION CONFERENCE (DAC)

Lee, Kyeongho; Cheon, Sungsoo; Jo, Joongho; Choi, Woong; Park, Jongsun

被引用次数: 14

2021

Sensory Adaptation and Neuromorphic Phototransistors Based on CsPb(Br_{1-x}I_x)₃ Perovskite and MoS₂ Hybrid Structure

ACS NANO

Hong, Seongin; Choi, Seung Hee; Park, Jongsun; Yoo, Hocheon; Oh, Joo Youn; Hwang, Euyheon; Yoon, Dae Ho; Kim, Sunkook

被引用次数: 112

2020

Rank order coding based spiking convolutional neural network architecture with energy-efficient membrane voltage updates

NEUROCOMPUTING

Tang, Hoyoung; Cho, Donghyeon; Lew, Dongwoo; Kim, Taehwan; Park, Jongsun

被引用次数: 15

2020

Domain Wall Memory-Based Design of Deep Neural Network Convolutional Layers

IEEE ACCESS

Chung, Jinil; Choi, Woong; Park, Jongsun; Ghosh, Swaroop

被引用次数: 5

2020

Special issue on SoC and AI processors

ETRI JOURNAL

Kim, Ji-Hoon; Lee, Minjae; Park, Jongsun; Cha, Ho-Young

被引用次数: 2

2020

An Energy-Quality Scalable STDP Based Sparse Coding Processor With On-Chip Learning Capability

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Kim, Heetak; Tang, Hoyoung; Choi, Woong; Park, Jongsun

被引用次数: 8

2020

Bit Parallel 6T SRAM In-memory Computing with Reconfigurable Bit-Precision

Design Automation Conference

Kyeongho Lee; Jinho Jeong; Sungsoo Cheon; Woong Choi; Jongsun Park

被引用次数: 0

2020

Prediction Confidence based Low Complexity Gradient Computation for Accelerating DNN Training

Design Automation Conference

Dongyeob Shin; Geonho Kim; Joongho Jo; Jongsun Park

被引用次数: 0

2020

Design of Processing-Inside"-Memory Optimized for DRAM Behaviors

IEEE ACCESS

Lee, Won Jun; Kim, Chang Hyun; Paik, Yoonah; Park, Jongsun; Park, Il; Kim, Seon Wook

被引用次数: 8

2019

Sensitivity-Based Error Resilient Techniques With Heterogeneous Multiply-Accumulate Unit for Voltage Scalable Deep Neural Network Accelerators

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Shin, Dongyeob; Choi, Wonseok; Park, Jongsun; Ghosh, Swaroop

被引用次数: 18

2019

A Perspective on Test Methodologies for Supervised Machine Learning Accelerators

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Motaman, Seyedhamidreza; Ghosh, Swaroop; Park, Jongsun

被引用次数: 9

2019

Charge-Recycling-Based Redundant Write Prevention Technique for Low-Power SOT-MRAM

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kang, Gyuseong; Park, Jongsun

被引用次数: 11

2019

Spike Counts Based Low Complexity SNN Architecture With Binary Synapse

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Tang, Hoyoung; Kim, Heetak; Kim, Hyeonseong; Park, Jongsun

被引用次数: 31

2019

Compact Implementations of HIGHT Block Cipher on IoT Platforms

SECURITY AND COMMUNICATION NETWORKS

Kim, Bohun; Cho, Junghoon; Choi, Byungjun; Park, Jongsun; Seo, Hwajeong

被引用次数: 15

2019

Area-Optimized Fully-Flexible BCH Decoder for Multiple GF Dimensions

IEEE ACCESS

Park, Byeonggil; Park, Jongsun; Lee, Youngjoo

被引用次数: 8

2018

Low Cost Convolutional Neural Network Accelerator Based on Bi-Directional Filtering and Bit-Width Reduction

IEEE ACCESS

Choi, Woong; Choi, Kyungrak; Park, Jongsun

被引用次数: 13

2018

Complementary logic operation based on electric-field controlled spin-orbit torques

NATURE ELECTRONICS

Baek, Seung-heon Chris; Park, Kyung-Woong; Kil, Deok-Sin; Jang, Yunho; Park, Jongsun; Lee, Kyung-Jin; Park, Byong-Guk

被引用次数: 110

2018

Energy Efficient Canny Edge Detector for Advanced Mobile Vision Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Lee, Juseong; Tang, Hoyoung; Park, Jongsun

被引用次数: 54

2018

A Low-Latency and Area-Efficient Gram-Schmidt-Based QRD Architecture for MIMO Receiver

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shin, Dongyeob; Park, Jongsun

被引用次数: 11

2018

Mosaic-CNN: A Combined Two-Step Zero Prediction Approach to Trade off Accuracy and Computation Energy in Convolutional Neural Networks

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Kim, Cheolhwan; Shin, Dongyeob; Kim, Bohun; Park, Jongsun

被引用次数: 16

2018

Analysis of Antenna Structure for Energy Beamforming in Wireless Power Transfer

IEEE TRANSACTIONS ON ANTENNAS AND PROPAGATION

Lim, Yongseok; Ahn, Hyun-Seok; Park, Jongsun

被引用次数: 14

2017

Improved Perturbation Vector Generation Method for Accurate SRAM Yield Estimation

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Choi, Woong; Park, Jongsun

被引用次数: 5

2017

Low-Complexity Massive MIMO Detectors Based on Richardson Method

ETRI JOURNAL

Kang, Byunggi; Yoon, Ji-Hwan; Park, Jongsun

被引用次数: 29

2017

Half-Select Free and Bit-Line Sharing 9T SRAM for Reliable Supply Voltage Scaling

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Shin, Kyungho; Choi, Woong; Park, Jongsun

被引用次数: 36

2017

Novel Folded-KES Architecture for High-Speed and Area-Efficient BCH Decoders

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Park, Byeonggil; An, Seungyong; Park, Jongsun; Lee, Youngjoo

被引用次数: 10

2017

Embedded DRAM-Based Memory Customization for Low-Cost FFT Processor Design

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kang, Gyuseong; Choi, Woong; Park, Jongsun

被引用次数: 11

2017

Adaptive ECC for Tailored Protection of Nanoscale Memory

IEEE DESIGN & TEST

Shin, Dongyeob; Park, Jongsun; Park, Jangwon; Paul, Somnath; Bhunia, Swarup

被引用次数: 5

2017

Mesostructured HfxAl_yO₂ Thin Films as Reliable and Robust Gate Dielectrics with Tunable Dielectric Constants for High-Performance Graphene-Based Transistors

ACS NANO

Lee, Yunseong; Jeon, Woojin; Cho, Yeonchoo; Lee, Min-Hyun; Jeong, Seong-Jun; Park, Jongsun; Park, Seongjun

被引用次数: 23

2016

Low-Power Channel-Adaptive Reconfigurable 4x4 QRM-MLD MIMO Detector

ETRI JOURNAL

Kumiawan, Iput Heri; Yoon, Ji-Hwan; Kim, Jong-Kook; Park, Jongsun

被引用次数: 0

2016

A Charge-Recycling Assist Technique for Reliable and Low Power SRAM Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Choi, Woong; Park, Jongsun

被引用次数: 14

2016

Exploiting Serial Access and Asymmetric Read/Write of Domain Wall Memory for Area and Energy-Efficient Digital Signal Processor Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Chung, Jinil; Ramclam, Kenneth; Park, Jongsun; Ghosh, Swaroop

被引用次数: 13

2016

Overview of Circuits, Systems, and Applications of Spintronics

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Ghosh, Swaroop; Iyengar, Anirudh; Motaman, Seyedhamidreza; Govindaraj, Rekha; Jang, Jae-Won; Chung, Jinil; Park, Jongsun; Li, Xin; Joshi, Rajiv; Somasekhar, Dinesh

被引用次数: 8

2016

Unequal-Error-Protection Error Correction Codes for the Embedded Memories in Digital Signal Processors

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Tang, Hoyoung; Park, Jongsun

被引用次数: 3

2016

A 0.4-mW, 4.7-ps Resolution Single-Loop $\Delta\Sigma$ TDC Using a Half-Delay Time Integrator

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Kwon, Chan-Keun; Kim, Hoonki; Park, Jongsun; Kim, Soo-Won

被引用次数: 8

2016

D2ART: Direct Data Accessing from Passive RFID Tag for infra-less, contact-less, and battery-less pervasive computing

MICROPROCESSORS AND MICROSYSTEMS

Lee, Joon Goo; Kim, Seon Wook; Kim, Dong-Hyun; Cho, Younga; Rieh, Jae-Sung; Kang, Gyusung; Park, Jongsun; Lee, Hokyu; Park, Sejin; Kim, Chulwoo

被引用次数: 2

2015

A Refresh-Less eDRAM Macro With Embedded Voltage Reference and Selective Read for an Area and Power Efficient Viterbi Decoder

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Choi, Woong; Kang, Gyuseong; Park, Jongsun

被引用次数: 21

2015

A Novel Phase-Control-Based Energy Beamforming Techniques in Nonradiative Wireless Power Transfer

IEEE TRANSACTIONS ON POWER ELECTRONICS

Lim, Yongseok; Park, Jongsun

被引用次数: 48

2015

Program Cache Busy Time Control Method for Reducing Peak Current Consumption of NAND Flash Memory in SSD Applications

ETRI JOURNAL

Park, Se-Chun; Kim, You-Sung; Cho, Ho-Youb; Choi, Sung-Dae; Yoon, Mi-Sun; Kim, Tae-Yun; Park, Kun-Woo; Park, Jongsun; Kim, Soo-Won

被引用次数: 0

2014

An Adaptive Impedance-Matching Network Based on a Novel Capacitor Matrix for Wireless Power Transfer

IEEE TRANSACTIONS ON POWER ELECTRONICS

Lim, Yongseok; Tang, Hoyoung; Lim, Seungok; Park, Jongsun

被引用次数: 425

2014

VL-ECC: Variable Data-Length Error Correction Code for Embedded Memory in DSP Applications

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Park, Jangwon; Park, Jongsun; Bhunia, Swarup

被引用次数: 15

2014

Improving Energy Efficiency in FPGA Through Judicious Mapping of Computation to Embedded Memory Blocks

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ghosh, Anandaroop; Paul, Somnath; Park, Jongsun; Bhunia, Swarup

被引用次数: 8

2014

Reconfigurable CORDIC-Based Low-Power DCT Architecture Based on Data Priority

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lee, Min-Woo; Yoon, Ji-Hwan; Park, Jongsun

被引用次数: 26

2014

An Efficient Memory-Address Remapping Technique for High-Throughput QC-LDPC Decoder

CIRCUITS SYSTEMS AND SIGNAL PROCESSING

Yoon, Ji-Hwan; Park, Jongsun

被引用次数: 1

2014

Flexible nano-hybrid inverter based on inkjet-printed organic and 2D multilayer MoS₂ thin film transistor

ORGANIC ELECTRONICS

Chung, Jong Won; Ko, Yeong Hwan; Hong, Young Ki; Song, Wongeon; Jung, Chulseung; Tang, Hoyoung; Lee, Jiyoul; Lee, Min Hyung; Lee, Bang-lin; Park, Jeong-il; Jin, Yongwan; Lee, Sangyoon; Yu, Jae Su; Park, Jongsun; Kim, Sunkook

被引用次数: 14

2014

Priority Based Error Correction Code (ECC) for the Embedded SRAM Memories in H.264 System

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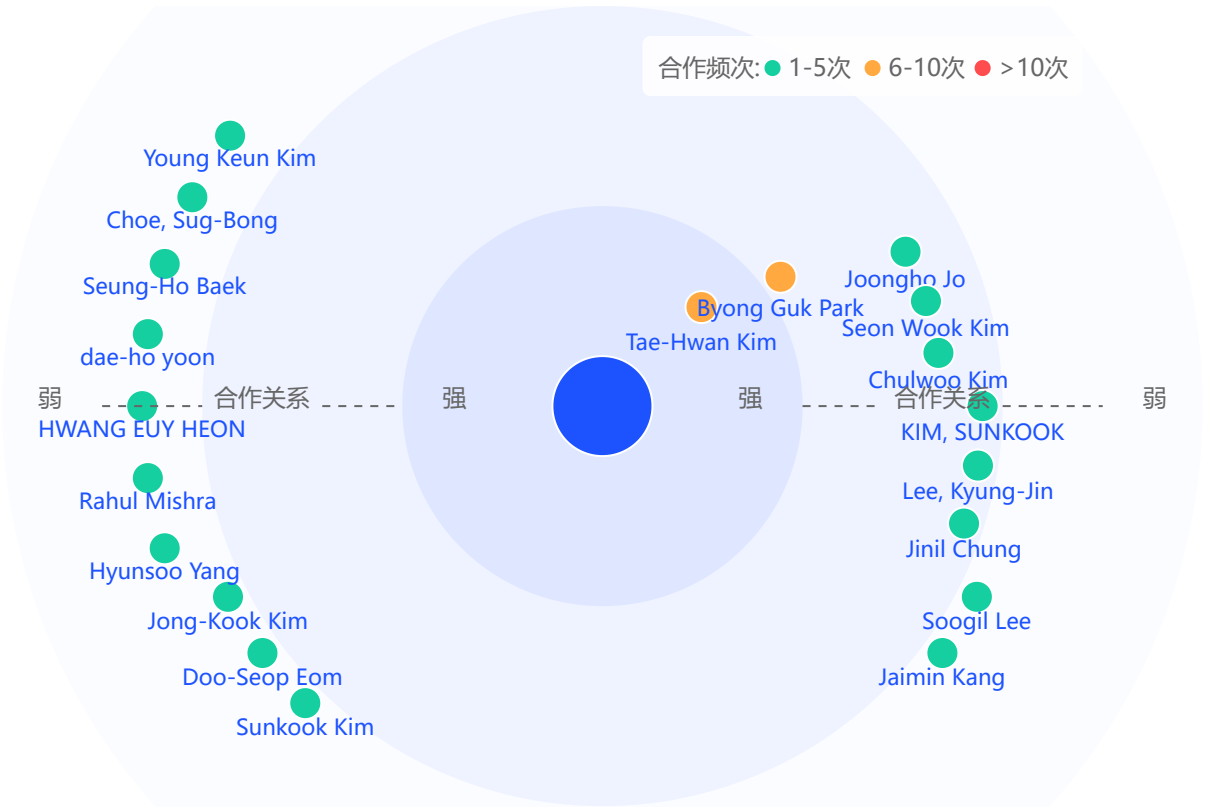
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